LEVELS OF SCIENTIFIC CONFIDENCE AND LOCATIONAL ACCURACY

Discrete levels of scientific confidence and locational accuracy have been developed to use as a terminology that can clearly yet concisely communicate the identity, existence, locatability, and positioning of geologic map features. These levels are directly derived from, or are closely associated with, the feature attributes (for more information, see Figures 1 and 2 in the accompanying introductory text). The following diagram shows how the various levels of scientific confidence and locational accuracy relate to the feature attributes; it also shows examples (in italics) of geologic situations to which the levels may be applied.

SCIENTIFIC CONFIDENCE		
Feature Attributes for Scientific Confidence		Levels of Scientific Confidence
Identity: • certain • questionable	Existence: • certain • questionable	Identity and existence certain *("I am certain that the planar feature I see in this outcrop is a fault")* Identity or existence questionable *("I can see some kind of planar feature in this outcrop, but I cannot be certain if it is a contact or a fault")*
LOCATIONAL ACCURACY		
Feature Attributes for Locational Accuracy		Levels of Locational Accuracy
Locatability: • observable • inferred • concealed	Zone of Confidence: • [value, in ground units] • [unit of measurement] Positioning Confidence: • within zone of confidence • may not be within zone of confidence	Location accurate *("I can clearly see this contact in outcrop, and I can accurately plot its position on the map")* Location approximate *("I can see this contact in outcrop, but I can't tell exactly where it is located because I am surrounded by trees")* Location inferred *("I can see by the change in debris materials visible around these gopher holes that a contact runs through here, but I can't locate it precisely")* Location concealed *("I can see that a contact is present on both sides of this lake, but I can't tell where it is located beneath the water")*

The levels of scientific confidence and locational accuracy have been used to identify and describe the line symbols in this standard. The following example is extracted from Appendix A (see p. A-1-1).

DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL
Contact—Identity and existence certain, location accurate	————————	Contact—Identity and existence certain, location inferred	_ _ _ _ _ _ _ _ _ _
Contact—Identity or existence questionable, location accurate	————?————	Contact—Identity or existence questionable, location inferred	_ _ _ _ _?_ _ _ _ _
Contact—Identity and existence certain, location approximate	— — — — — —	Contact—Identity and existence certain, location concealed	- - - - - - - - - - - - - - - - -
Contact—Identity or existence questionable, location approximate	— — —?— — —	Contact—Identity or existence questionable, location concealed	- - - - - - - - ? - - - - - - - -

In the symbol explanation on a published map, the levels of scientific confidence and locational accuracy also are used to identify and describe the various types and styles of line symbols that appear on the map; however, not every style of a particular line symbol needs to be listed individually in the explanation, as the following two examples show.

Contact—Solid where location is accurate; long-dashed where location is approximate; short-dashed where location is inferred; dotted where location is concealed. Queries added where identity or existence may be questionable

Fault—Solid where location is accurate; long-dashed where location is approximate; short-dashed where location is inferred; dotted where location is concealed. Queries added where identity or existence may be questionable

Thrust fault—Solid where location is accurate; long-dashed where location is approximate; short-dashed where location is inferred; dotted where location is concealed. Queries added where identity or existence may be questionable. Sawteeth on upper plate

Folds—Solid where location is accurate; long-dashed where location is approximate; dotted where location is concealed. Queries added where identity or existence may be questionable. Showing direction of plunge where appropriate

Anticline

Syncline

The example on the left shows how the descriptions of all styles of a particular line type are grouped into one explanatory paragraph (if the symbol explanation directly follows the Description of Map Units, a separate title is not added).

The example below shows a more condensed alternative in which the descriptions of all line-symbol styles are summarized in a bracketed headnote (a separate title usually is added).

SYMBOL EXPLANATION

[For all line symbols: lines are solid where location is accurate; long-dashed where location is approximate; short-dashed where location is inferred; dotted where location is concealed. Queries added where identity or existence may be questionable]

Contact

Fault

Thrust fault—Sawteeth on upper plate

Folds—Showing direction of plunge where appropriate

Anticline

Syncline